

Kazi Md. Tawsif Rahman

Dhaka, Bangladesh | tawsifcse113@gmail.com | +880 1533-406937 | linkedin.com/in/tawsifrahman113
github.com/tawsif113 | **Research portfolio:** research.tawsifrahman.flaro-tech.com

Research Summary

Computer Science graduate and backend systems engineer researching differential privacy and membership-inference risk in machine-learning-based network intrusion detection. My current independent study uses Opacus-based DP-SGD, explicit (epsilon, delta)-privacy accounting, and shadow-model membership-inference auditing under a locked evaluation protocol. I completed the five-seed NSL-KDD analysis and a constrained single-seed UNSW-NB15 external validation. The experimental roadmap is frozen and a venue-neutral first paper draft is under author review.

Research Interests

AI security; differential privacy; privacy-preserving machine learning; membership-inference auditing; trustworthy evaluation of security-critical ML; federated learning for security applications; reproducible experimental systems.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh	July 2024
B.Sc. in Computer Science and Engineering	CGPA: 3.59 / 4.00

Research Experience

Privacy-Utility Auditing of DP-SGD for ML-Based Network Intrusion Detection	2026 - Present
<i>Independent Research Project</i>	
- Designed a locked NSL-KDD binary-classification protocol with a 70/10/20 target-train/validation/shadow split, validation-only F2 threshold selection, and untouched KDDTest+ utility evaluation; RF, XGBoost, and MLP utility baselines are retained as archived context. - Built five shadow MLPs and evaluated score-only and label-aware membership-inference attacks. The strongest evaluated non-private attack remained near chance, creating a floor effect reported as an evaluation limitation rather than evidence of privacy. - Integrated Opacus DP-SGD with explicit privacy accounting, clipping, sampling, noise, and training parameters recorded for reproducibility. - Completed five target-training seeds for non-private, epsilon about 4, and epsilon about 2 on NSL-KDD. Mean Recall was 70.93%, 71.34%, and 70.11%; the epsilon-about-4 Recall difference was uncertain, while false-positive rate increased and average precision decreased. - Paired membership-inference AUC intervals crossed zero, so the accepted NSL-KDD evidence does not support a claim that DP-SGD reduced measured overall membership leakage. - Completed a constrained single-seed UNSW-NB15 validation. Private settings preserved selected-threshold F1 and Recall closely but increased false-positive burden and reduced average precision; overall MIA remained near chance. - The experimental roadmap is complete. A venue-neutral first manuscript is frozen for author review and target-venue selection; no tested epsilon is claimed to be universally optimal.	
<i>Code, results, and reproducibility artifacts:</i> github.com/tawsif113/privacy-utility-dp-ids	

Publications

- Kazi Md. Tawsif Rahman** and Chowdhury Mahfuzul Hoq. “An Automated System for Detecting Property Insurance Fraud Using Machine Learning.” *International Journal of Mathematical Sciences and Computing*, 2024. doi:10.5815/ijmsc.2024.03.02.
- Mahfuzulhoq Chowdhury, **Kazi Md. Tawsif Rahman**, and Hossain Ahmad Maruf. “Whistle Blower: An Insurance Awareness Mobile Application with Insurance Policy Selection, Fraud Detection, Critical Help, Complaint Features.” *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, pp. 1-6, 2024. doi:10.1109/CSITSS64042.2024.10817002.

Selected Professional Experience

Software Engineer, BRAC IT Services Ltd., Dhaka, Bangladesh	Dec 2024 - Present
CRM Platform	
- Built configurable rule-based routing with priorities, schedules, conditional matching, rotation rules, duplicate checks, and validated field/operator/value conditions. - Replaced full-column Kanban reordering with fixed-length order keys and reduced each movement to a single-rank assignment. - Added access controls, idempotent moves, transition validation, audit logs, and timeline entries to make state changes traceable.	
Ticketing System	
- Built services for ticket creation, search, comments, forwarding, assignment rules, teams, SLA/KPI tracking, feedback, notifications, and metadata. - Implemented least-loaded team routing with Redis-backed workload counters and maintained PostgreSQL-Redis workload consistency after ticket creation, reassignment, and resolution.	

Technical and Research Skills

Research methods	Locked-split experimental design, validation-only threshold selection, shadow-model membership-inference auditing, privacy accounting, paired bootstrap confidence intervals, reproducibility manifests, threat-model specification
ML / Privacy	Python, scikit-learn, XGBoost, PyTorch, Opacus, DP-SGD, membership-inference attacks, model evaluation

Backend / Data	Java, Spring Boot, Spring Security, JPA/Hibernate, REST APIs, PostgreSQL, MongoDB, Redis, RabbitMQ, Flyway
Engineering	SQL, C, C++, Docker, Linux, Git, JUnit 5, Mockito, API testing

Awards and Achievements

- Codeforces Specialist; rating above 1500.
- Finalist, CUET IUPC 2024 and CUET IUPC 2023.
- Finalist, CUET Tech Carnival 2020.

Selected Coursework

Algorithms and Data Structures; Database Systems; Computer Networks; Operating Systems; Software Engineering; Artificial Intelligence; Machine Learning; Information Security.